

Recap

Unintentional data leakages are under the responsibility of the users who tend to use weak/short passwords, overshare and click links without paying sufficient attention

Intentional data leakages are the result of security attacks

These attacks are often successful because of the user behaviors

Data leakages might have serious impacts on individuals and are generally very expensive for companies

Necessary to protect data and infrastructure

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Provider's perspective

To protect data and infrastructure, providers should reduce security risks and their attack surface

Preventive measures:

- Regular application of security *patches* to all software and devices
- Secure network *configuration*
- Regular *data backups*, stored independently and appropriately protected
- Policies to *manage* and *monitor* access to sensitive data, including audit trails for changes to records
- Business continuity plans (especially for critical infrastructure and services)

Security mechanisms that ensure data confidentiality and integrity and mitigate the risks of data breaches

- Encryption
- Access control

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Encryption

Encryption is the process of converting readable data (*plaintext*) into an unreadable format (*ciphertext*) using a mathematical algorithm and a secret key

Only authorized parties with the correct decryption key can reverse the process and read the original data

Encryption is strong measure

Encryption can be applied to

- *Data in transit*, i.e., data transferred over networks
- *Data at rest*, i.e., inactive data stored on a device

Encrypted data can be *safely* stored in the cloud or on mobile devices, thus ensuring data remains “secure” and accessible anytime from anywhere

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Access control mechanisms

Access control mechanisms refer to the techniques and processes used to control and manage the access to data and resources within a system

Access to data/resources should be restricted to “authorized” users by specifying *who* can access what, under *what conditions*, and with what *level of privileges*

Authentication is an *explicit* access control mechanism aimed at verifying the identity of a user (e.g., credentials, pins, biometric data)

Authorization is an *implicit* mechanism that defines the actions and resources of an authenticated user by granting or denying access permissions based on the user identity and defined roles

Authorization mechanisms are also based on device properties (e.g., MAC address, IP address)

Sophisticated mechanisms, e.g., Role-Based Access Control, Access Control Lists, for managing access permissions

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Access control: best practices

Effective authentication and authorization mechanisms are crucial for ensuring integrity of data, systems and services

Best practices include:

- Password enforcement mechanisms
- Multi-factor authentication (MFA) with One Time Password (OTP) & biometric data (2FA special case)
- Regular review and updates of user roles and access permissions
- Enforcement of the principle of *least privilege*, granting users only the *necessary* access permissions

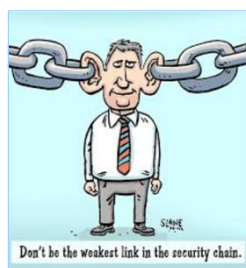


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Yet another security measure



Individuals are often the weakest link of the security chain and are chronically responsible for the failure of security systems

To ensure security and privacy a very important measure is

Education

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General Data Protection Regulation

General Data Protection Regulation (GDPR) is a general framework for data protection laws in the European Union (EU) effective since May 25, 2018

It provides rules on the protection of natural persons with regard to processing personal data (PII) and on the free movement of such data

It applies to EU citizens and to any company that processes their data

It provides individuals with strong control over personal data

The regulation identifies two main actors:

- A *data controller* (e.g., our Rector) who defines the purposes and means of processing personal data, that is, why and how data is processed
- A *data processor* (e.g., employee, third party) who processes the data on behalf of the data controller

Rules refer to the obligations of data controllers and data processors

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GDPR obligations

Provide detailed information about which data is being collected and how it is used, stored, shared, transferred, ...

Implement mechanisms to obtain an *explicit consent*, that is, “*freely given, specific, informed and unambiguous*” of the users

Implement mechanisms to allow users to access, delete or modify their data

Protect data being collected and processed from unauthorized access and misuse with appropriate data security measures

Notify users in the event of a data breach (72 hours)



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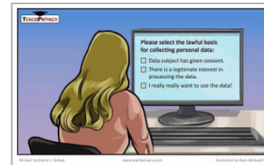
GDPR compliance

GDPR compliance is a *legal* requirement

Data controllers hold the primary responsibility for legal compliance, ensuring that data is processed securely, respecting user rights, and managing data processors

Data Protection Impact Assessment (DPIA) is recommended to check the compliance

Third party services must be included in the DPIA



Companies that violate GDPR privacy and security standards risk severe penalties, including fines of up to €20 million or 4% of their annual revenue, whichever is higher

451: Unavailable due to legal reasons

We recognize you are attempting to access this website from a country belonging to the European Economic Area (EEA) including the EU which enforces the [General Data Protection Regulation](#) (GDPR) and therefore access cannot be granted at this time. For any issues, contact webmaster@uniondemocrat.com or call [209-532-7151](tel:209-532-7151).

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European data protection for the digital era



Better protection for personal data



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GDPR violations in the news

Meta hit with record \$1.3 bln fine over data transfers

By Padraic Halpin
May 22, 2023 4:43 PM GMT+2 · Updated May 22, 2023

Facebook owner Meta fined €1.2bn for mishandling user information

Penalty from Ireland's privacy regulator is a record for breach of EU data protection regulation

TikTok fined 530 million euros by EU regulator over data protection

By Reuters
May 2, 2025 11:15 PM GMT+2 · Updated May 2, 2025

€530m TikTok fine highlights the importance of effective international data transfer mechanisms

Published on 23 September 2025

Italy's Garante fines Enel Energia over €79m for data privacy violations in telemarketing

A record fine of over €79m (\$85.4m) has been imposed on [Enel Energia](#) over alleged mishandling of numerous electricity and gas supply customers' personal data for telemarketing purposes.

Reuters

Italy fines OpenAI over ChatGPT privacy rules breach

By Elvira Polina and Alvise Armellini
December 20, 2024 9:51 PM GMT+1 · Updated 3 months ago

Italy's privacy watchdog fines OpenAI €15 million after probe into ChatGPT data collection

Company says decision disproportionate, to file appeal

The fine comes after the authority found OpenAI processed users' personal data to "train ChatGPT without having an adequate legal basis and violated the principle of transparency and the related information obligations towards users".

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Largest fines for GDPR violations

Company (Date, Location)	Fine (million euros)
Meta Platforms, Inc. (May 2023, Ireland)	1,200
Amazon Europe Core S.à.r.l (Jul 2021, Luxembourg)	746
TikTok Technology Limited (May 2025, Ireland)	530
Meta Platforms, Inc. (Sep 2022, Ireland)	405
Meta Platforms Ireland Ltd. (Jan 2023, Ireland)	390
TikTok Ltd. (Sep 2023, Ireland)	345
LinkedIn (Oct 2024, Ireland)	310
Uber Technologies Inc., Uber B.V. (Jul 2024, Netherlands)	290
Meta Platforms Ireland Ltd. (Nov 2022, Ireland)	265
WhatsApp Ireland Ltd. (Dec. 2024, Ireland)	251

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GDPR violations outcomes

WhatsApp fined a record 225 mln euro by Ireland over privacy

By Conor Humphries

September 2, 2021 3:46 PM GMT+2 · Updated September 2, 2021



WhatsApp wins court backing to challenge \$268 million Irish privacy fine

By Foo Yun Chee and Padraic Halpin

February 10, 2026 9:56 AM GMT+1 · Updated February 10, 2026

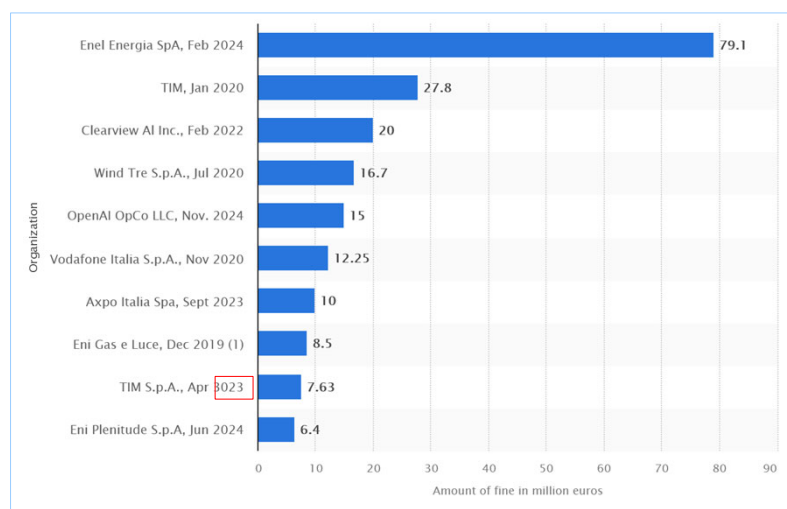


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Largest fines for GDPR violations in Italy



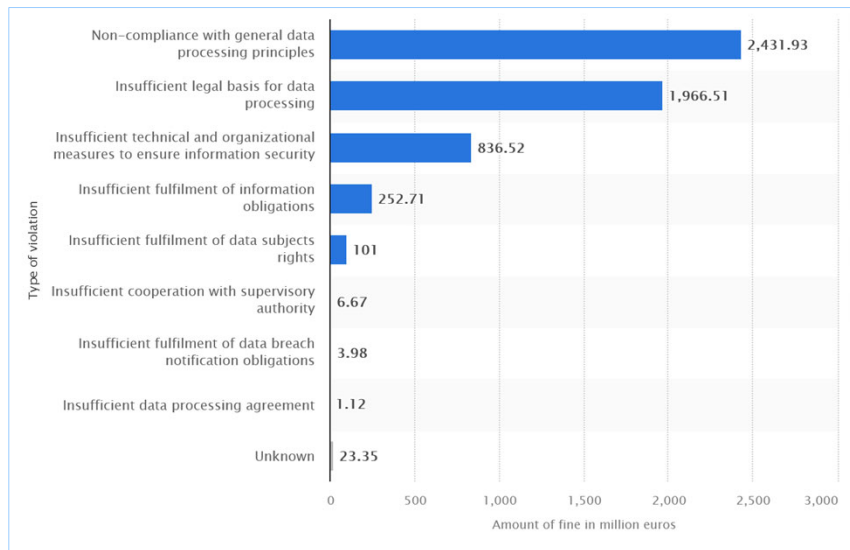
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Fines by violation type



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European legislation on cybersecurity



The [Cybersecurity Act](#) [adopted in 2019] aims at improving protection against security threats within the European Union by establishing an EU-wide cybersecurity *certification framework* for ICT products, services and processes

Companies will be able to demonstrate that they have adopted security best practices



The [Cyber Resilience Act](#) aims at enhancing the transparency of the security properties and promoting trust by introducing *mandatory* cybersecurity requirements for *products that include digital elements*

Obligations will apply from December 11, 2027



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Impacts of Cybersecurity

Security must be integrated in the entire lifecycle of digital products to comply with legislation and avoid incidents due to vulnerabilities (*security by design*)

Direct costs of security refer to the investments in terms of human resources and technological infrastructure related to the design and development of “secure” products and the prevention, detection and mitigation of security threats

Indirect costs of security refer to the costs related to business and reputation losses due to data breaches and security incidents

Setup of an appropriate *incident response plan*, i.e., a set of instructions with the specification of the role and responsibility of the security team, to help security team detect, respond to, and recover from security incidents

All these requirements increase the overall cost associated with services/infrastructures

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Dependability: summary

Availability: readiness

- » scheduled/unscheduled downtimes affect availability
- » downtimes are due to maintenance activities, human errors, bugs, misconfigurations, security attacks & reliability of technological infrastructure

Reliability: continuity

- » robustness of components
- » appropriate redundancy, replication, distribution & diversity

Trustworthiness:

- » security & privacy, confidentiality, integrity & availability
- » security threats affect infrastructure/services and data
- » measures to protect infrastructure/services, prevent security incidents and promptly react in case of incidents
- » measures to educate users & create awareness towards security principles

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Performance

Performance is a quality attribute that measures the *efficiency* of a service or technological infrastructure in terms of *speed* and *time* to complete a given task

Performance is often perceived as a synonym of quality

Very subjective attribute

Performance affects usability

Two metrics are used to measure performance:

- *Throughput*, i.e., amount of work performed per time unit (e.g., bandwidth, flops, fps)
→ it should be **maximum**
- *Time*, i.e., how long it takes to complete a given task (e.g., network latency, page load time, query time)
→ it should be **minimum**

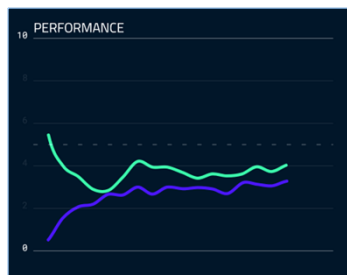
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Performance (cont'd)

It is difficult to ensure “good” performance levels



Actual performance affected by the characteristics and intensity of the workload/traffic (e.g., nominal vs actual bandwidth, on demand services, social nature of services, virtualized resources)

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Service Level Agreement

A Service Level Agreement is a **contract** between an external or internal provider and its customers

The contract is a *legally binding* document that specifies *technical, legal, administrative* and *financial* provisions about the service

These *provisions* refer to:

Legal Stuff

- *conditions* under which the service is delivered by the provider
- *responsibilities* (i.e., rights, duties) of the customers
- *service levels* to be expected

The contract must ensure that both parties have the same understanding of the agreed provisions

No party can plead ignorance!



Contracts vary with the type of service (e.g., premium, free), type of customers (e.g., business, consumer) and cost associated with the service

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SLA Provisions

The contract is a *live document* that includes:

- Description of the *service* (i.e., what service is about and what provider offers)
- Description of the *service levels* (i.e., how provider will implement the offer, expressed in terms of *quantitative/measurable* quality indicators)
- Description of the service level *monitoring & reporting* (i.e., who, how and how often the indicators will be measured and made available to customers)
- Description of consequences for *violations* of service obligations (i.e., penalties, indemnifications, contract termination); force-majeure excluded
- Description of *customer responsibilities* (i.e., what customers can do or must not do)
- Description of *contract evolution* (i.e., renewal, duration, changes)

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Example of SLA template



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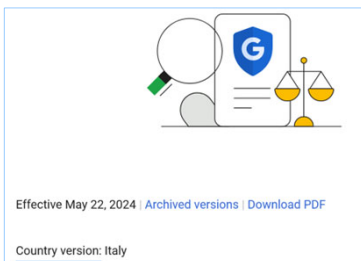
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Google Terms of Service



Effective May 22, 2024 | [Archived versions](#) | [Download PDF](#)

Country version: [Italy](#)

What's covered in these terms

We know it's tempting to skip these Terms of Service, but it's important to establish what you can expect from us as you use Google [services](#)®, and what we expect from you.

[Google Terms of Service](#)

These Terms of Service reflect [the way Google's business works](#) [↗](#), the laws that apply to our company, and [certain things we've always believed to be true](#) [↗](#). As a result, these Terms of Service help define Google's relationship with you as you interact with our services. For example, these terms include the following topic headings:

- [What you can expect from us](#), which describes how we provide and develop our services
- [What we expect from you](#), which establishes certain rules for using our services
- [Content in Google services](#), which describes the intellectual property rights to the content you find in our services — whether that content belongs to you, Google, or others
- [In case of problems or disagreements](#), which describes other legal rights you have, and what to expect in case someone violates these terms

Understanding these terms is important because, to use our services, you must accept these terms.

Besides these terms, we also publish a [Privacy Policy](#). Although it's not part of these terms, we encourage you to read it to better understand how you can [update, manage, export, and delete your information](#) [↗](#).

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Google Privacy Policy



GOOGLE PRIVACY POLICY

When you use our services, you're trusting us with your information. We understand this is a big responsibility and work hard to protect your information and put you in control.

This Privacy Policy is meant to help you understand what information we collect, why we collect it, and how you can update, manage, export, and delete your information.

If European Union or United Kingdom data protection law applies to the processing of your information, you can review the [European requirements section](#) below to learn more about your rights and Google's compliance with these laws.

 **Privacy Checkup**
Looking to change your privacy settings?
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Google Privacy & Terms

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[Information Google collects](#)

[Why Google collects data](#)

[Your privacy controls](#)

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Google SLAs

Google Workspace Service Level Agreement

Last modified: January 30, 2025

"Downtime" means, for a domain, a period of time during which the user web interface for the applicable Covered Services used by Customer has more than a five percent user error rate. Downtime is measured based on server side error rate.

"Monthly Uptime Percentage" means total number of minutes in a calendar month minus the number of minutes of Downtime in a calendar month, divided by the total number of minutes in a calendar month.

Monthly Uptime Percentage	Days of Service added to the end of the Service term (for offline billing customers), or monetary credit equal to the value of days applied to a future invoice (for online billing customers), for the applicable Covered Service(s) that did not meet the Google Workspace SLA
< 99.9% - >= 99.0%	3
< 99.0% - >= 95.0%	7
< 95.0%	15

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AWS SLAs

AWS Service Level Agreements (SLAs)

AWS commits to offer Service Level Agreements (SLAs) for all paid, generally available services.

AMAZON ELASTIC COMPUTE CLOUD (EC2) (INSTANCE LEVEL)

Instance-Level SLA

For each individual Amazon EC2 instance ("Single EC2 Instance"), AWS will use commercially reasonable efforts to make the Single EC2 Instance available with an Instance-Level Uptime Percentage of at least 99.5%, in each case during any monthly billing cycle (the "Instance-Level SLA"). In the event any Single EC2 Instance does not meet the Instance-Level SLA, you will be eligible to receive a Service Credit as described below.

Instance-Level Uptime Percentage	Service Credit Percentage
Less than 99.5% but equal to or greater than 99.0%	10%
Less than 99.0% but equal to or greater than 95.0%	30%
Less than 95.0%	100%

Note: In addition to the Instance-Level SLA, AWS will not charge you for any Single EC2 Instance that is Unavailable for more than six minutes of a clockhour. This applies automatically and you do not need to request credit for any such hour with more than six minutes of Unavailability.

Amazon S3 Service Level Agreement

Last Updated: November 28, 2023

Monthly Uptime Percentage	Service Credit Percentage
Less than 99.9% but greater than or equal to 99.0%	10%
Less than 99.0% but greater than or equal to 95.0%	25%
Less than 95.0%	100%

Definitions

- "Error Rate" means, for each request type to an Amazon S3 storage class: (i) the total number of internal server errors returned by Amazon S3 for such request type to the applicable Amazon S3 storage class as error status "InternalError" or "ServiceUnavailable" divided by (ii) the total number of requests for such request type during the applicable 5-minute interval of the monthly billing cycle. We will calculate the Error Rate for each AWS account as a percentage for each 5-minute interval in the monthly billing cycle. The calculation of the number of internal server errors will not include errors that arise directly or indirectly as a result of any of the Amazon S3 SLA Exclusions.
- "Monthly Uptime Percentage" is calculated by subtracting from 100% the average of the Error Rates from each 5-minute interval in the monthly billing cycle. If you did not make any requests in a given 5-minute interval, that interval is assumed to have a 0% Error Rate.

Amazon S3 SLA Exclusions

The Service Commitment does not apply to any unavailability, suspension or termination of Amazon S3, or any other Amazon S3 performance issues: (i) caused by factors outside of our reasonable control, including any force majeure event or Internet access or related problems beyond the demarcation point of Amazon S3; (ii) that result from any actions or inactions of you; (iii) that result from your equipment, software or other technology; or (iv) arising from our suspension or termination of your right to use Amazon S3 in accordance with the AWS Agreement (collectively, the "Amazon S3 SLA Exclusions"). If availability is impacted by factors other than those used in our calculation of the Monthly Uptime Percentage, then we may issue a Service Credit considering such factors at our discretion.

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Microsoft SLAs

Service Level Agreements (SLA) for Online Services

The Service Level Agreements (SLA) describe Microsoft's commitments for uptime and connectivity for Microsoft Online Services. The current and archived editions of the SLA are available for download and they cover Azure, Dynamics 365, Office 365, and Intune.

Microsoft 365 Apps for business

Downtime: Any period of time when Office applications are put into reduced functionality mode due to an issue with Office 365 activation.

Uptime Percentage: The Uptime Percentage is calculated using the following formula:

$$\frac{\text{User Minutes} - \text{Downtime}}{\text{User Minutes}} \times 100$$

where Downtime is measured in user-minutes; that is, for each Applicable Period, Downtime is the sum of the length (in minutes) of each Incident that occurs during that Applicable Period multiplied by the number of users impacted by that Incident.

Service Credit:

Uptime Percentage	Service Credit
< 99.9%	25%
< 99%	50%
< 95%	100%

[Table of Contents / Definitions](#)

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BTnet – Service Level Agreement

Network latency

1. We agree to provide your service with a latency guarantee subject to the terms of this agreement.
2. We'll work out this latency by calculating the average of round-trip transmission measurements taken in ten minute intervals during a calendar month. The results will be posted on www.bt.net.
3. The latency guarantee depends on the network you're using:
The UK core network connection: The latency guarantee is an average round-trip transmission time of 20 milliseconds or less between BT selected core Point of Presence (PoP) in the BT network.
The transatlantic network connection: The latency guarantee is an average round-trip transmission time of 95 milliseconds or less between designated transit routers at each end of the transatlantic link.
4. If we don't meet the latency guarantee for two consecutive months we'll give you a discount. You can save the equivalent to one day's rental charge for your service.
5. You don't need to make a claim, we'll just automatically apply the reduction to your rental charges.

Service availability

1. We aim to provide your service with a target of 100% availability at all times, subject to the terms of this agreement.
2. If there's evidence been an outage, we'll reduce your BTnet rental charge for the service as follows:
 - Where the outage period lasts for ten hours or less, you'll save the equivalent of one day's rental charge per hour of downtime. When calculating the outage period, we'll round up the total time to the nearest hour.
 - The maximum reduction applied per quarter will be capped at ten hours of outage, which is equivalent to ten days' rental charge for your service.
3. You should report an outage within two days by calling 0808 100 2440. We'll provide you with a trouble ticket number for your reference.
4. Once we've verified the outage, you'll need to make a claim within 28 days, quoting the relevant trouble ticket number, customer reference (FTIP number) and reason for the claim by email to: btnet.slg@bt.com

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Acceptable Use Policy

The BT IP Network is a resource from which we provide a variety of services to our customers. Our objective in managing this network is to:

- Provide a high quality, reliable service to our customers
- Respect the privacy of our customers
- Protect the security/integrity of our network and systems

Prohibited Uses

Examples of prohibited uses of the BT IP Network are described below. The examples are guidelines and are not intended to be exhaustive.

Illegal/Criminal Activity - The BT IP Network may not be used in connection with criminal or civil violations of laws, regulations, or other government requirements of any jurisdiction. Such violations include theft or infringement of copyrights, patents, trademarks, trade secrets, or other intellectual property; export control violations; fraud; forgery; pyramid or other prohibited business schemes; and theft, misappropriation, or unauthorised transmission or storage of funds, credit card information, personal information, or online services.

Security Violations - The BT IP Network may not be used to violate the security of a network, service, or other system. Examples of security violations include hacking, cracking into, monitoring, or using systems without authorisation; scanning ports; conducting denial of service attacks; distributing viruses or other harmful software; smurf attacks; and unauthorised alteration or destruction of web sites or other information.

Threats - The BT IP Network may not be used to transmit or store material of a threatening nature, including threats of death or physical harm, harassment, libel, and defamation.

Offensive Material - The BT IP Network may not be used to transmit or store material that is illegal according to the relevant national laws. Examples of material that might be subject to these laws may include:

- Material of an offensive nature, including obscene and pornographic material (particularly pornography involving children)
- Indecent, abusive, libelous and hateful materials

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Why performance varies

Poor performance and inefficiencies are due to load/traffic conditions or inadequate infrastructures and lead to SLA violations & financial/reputation losses

Unforeseen interactions between components (that perform well in isolation) might lead to poor performance

A failure of a component might overload other components (*cascade effects*), thus leading to poor performance (and unavailability)

It is necessary to identify precisely the causes of the performance problems

Bottleneck is defined as the component limiting the performance of a system

A component is not a bottleneck by definition

It might become a bottleneck under specific conditions



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Performance engineering approaches

Performance engineering is the discipline that studies the efficiency of systems/services (under design or being deployed) and the quality experienced by the users

Two types of performance engineering approaches:

- *reactive* approach whose objective is to find remedies as soon as problems appear by applying (temporary or permanent) *tuning* actions (i.e., “adjustments” of the behavior of an infrastructure to mitigate the symptoms and effects of the bottleneck)
- *proactive* approach whose objective is to predict and anticipate performance problems by applying *capacity planning* actions (i.e., analysis of the current conditions of an infrastructure, prediction of its residual capacity and of its evolution through a *what-if* analysis)

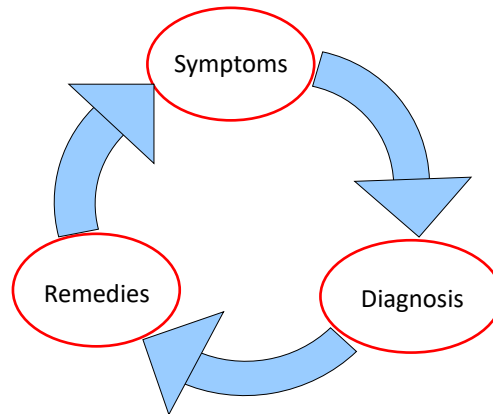
Both approaches should be applied (i.e., anticipation & reaction)

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How to identify & remove performance issues?



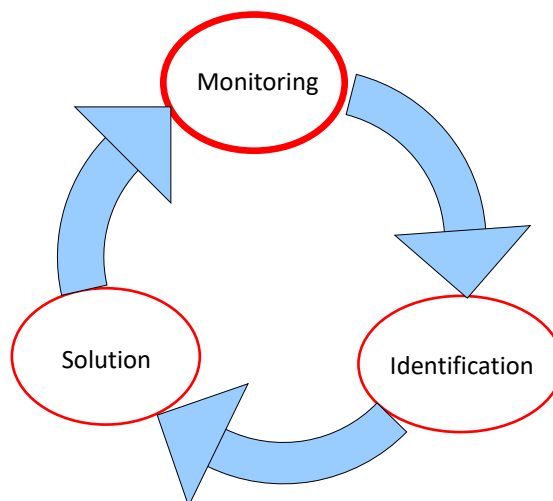
All these activities require a solid knowledge of the functionalities and behavior of systems/services (*domain knowledge*)

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Performance analysis in detail



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